Bench

1 GC B cells



Peyer's patches; sort B220^{+}GL7^{+}Fas^{+}, ~10,000 cells per mouse

steps 1–4 · 4–5 h

2 Genomic DNA



proteinase K lysis; Wt and AID-deficient mice side by side

steps 5–6 · 2 h

3 J_H4 PCR



1.2-kb amplicon, high-fidelity polymerase; gel check, clean-up

steps 7–10 · 3 h

4 Sequencing



full-length amplicons, R10.4.1 flow cells; FASTQ per library

steps 11–12 · 1–2 days

Analysis with SHMtrace

5 Project



sample sheet: file, sample, group, control; app or command line

steps 13–15 · 15 min

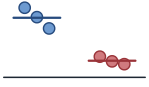
6 Run



read preparation, control-calibrated calling, animal-level tests

steps 16–17 · 2–10 min

7 Report



controls, margin, read accounting, warnings; spectra, maps, clones

steps 18–23 · 30 min

8 Plan



power from this analysis as a pilot: animals per group

step 24 · 5 min